LESSON 1.3

Math is In My World

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can represent a real-world situation with a tape diagram or table and use decimal operations to solve it.

WHY THIS MATTERS Math lives in your world — in coins on a table and tram cars over a valley. Today you visualize real situations, pick tools like tape diagrams and tables, and let decimal arithmetic do the heavy lifting. The tram must carry 4,000 passengers over the valley. Each ride carries 80 passengers and takes 12.5 minutes. How many rides are needed, and how long will all of the rides take?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
representation *Spanish: representación*	A picture, diagram, table, or equation that shows the mathematics in a situation.	The same 12 shown three ways: a tape diagram, a table, and 3 × 4.
tape diagram *Spanish: diagrama de cintas*	A rectangle cut into equal parts that helps you visualize how a total breaks apart.	One bar of 4,000 passengers cut into equal groups of 80.
tool *Spanish: herramienta*	Anything you choose to help you see relationships and solve a problem — a table, a diagram, a graph.	A table, a number line, or a tape diagram — whichever shows the relationship.
ordered pair *Spanish: par ordenado*	Two numbers that go together, like a time and a number of trips, that you can plot as a point.	(4, 60): 4 hours costs $60 — the hours always come first.
round trip *Spanish: viaje de ida y vuelta*	A complete journey from the start, to the destination, and back to the start.	5 miles out plus 5 miles back = 10 miles in all.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A pile of coins is too big to count one at a time. About how many coins are in the pile?

1. I do not count every coin — I find a small stack I CAN count, about 10 coins.
2. Then I estimate how many stacks like that are in the pile. I count about 15.
3. 15 stacks of about 10 is 15 × 10 = 150, so my estimate is about 150 coins — reasoned, not random.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

One tram ride takes 12.5 minutes, so 10 rides take 12.5 × 10 = 125 minutes and 50 rides take 12.5 × 50 = 625 minutes.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- This situation needs _____ because we are _____ a total, not _____ it.
- First I counted _____ .
- Then I estimated about _____ because _____ .
- My tape diagram shows _____ .

Word bank representation | tape diagram | tool | ordered pair | round trip

WATCH OUT!

A common mistake

Converting minutes to hours by moving the decimal point

LESSON 1.3

Math is In My World

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

One tram ride takes 12.5 minutes. How many minutes do 10 rides take?

(A) 1.25 minutes
(B) 22.5 minutes
(C) 125 minutes
(D) 1,250 minutes

STEPS TO TRY
1. You need 10 equal groups of 12.5.
2. Multiplying by 10 moves the decimal point one place.
3. Which direction makes the number bigger?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Five Breaks: which of these is a correct way to decompose 12.5?

(A) 1 + 2.5

(B) 10 + 2.5

(C) 10 + 2.05

(D) 12 + 5

STEPS TO TRY

1. Add each option and compare to 12.5.
2. What does the 5 in 12.5 stand for?
3. Add the two parts in each choice, and keep the tenths in mind — 12.5 is 12 and five tenths.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

How many minutes do 50 tram rides take, at 12.5 minutes per ride?

(A) 62.5 minutes

(B) 600 minutes

(C) 625 minutes

(D) 6,250 minutes

STEPS TO TRY

1. Use the table pattern: 10 rides take 125 minutes.
2. 50 rides is 5 groups of 10 rides.
3. 125 × 5 = ?

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

PART 2 Explain and correct

reasoning

4 **Answer in complete sentences.**

A coin display has about 15 stacks that look the same size. You count 10 coins in one stack. About how many coins are in the display? Explain how to estimate without counting every coin.

SENTENCE STARTER First I counted ___ .
Then I estimated about ___ because ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Find a card where you and your partner chose different operations. What is it about the situation — not the numbers — that tells you whether division or multiplication helps?

SENTENCE STARTER This situation needs ___ because we are ___ a total, not ___ it.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help